

Course Information

Course Number: VIST110
Course Title: Design Innovation
Time: 2 hours lecture and 2 hours lab
Location: TBD
Credit Hours: 3

Instructor Details

Instructor:	Michael Bruner	TA:
Phone:	N/A	E-Mail:
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Office Hours:	by appointment	

Catalog Course Description

Introduction to fundamental design issues, processes, and theories relevant to design resolution and the creation of new ideas for innovative design products and services by keeping human needs at the center of the development process; Overview of design research, prototyping, evaluation, and communication techniques to generate valuable insights for design innovation.

Course Introduction

This course allows students to envision and to generate new products, processes, services, and experiences of creative expression. Through individual and collaborative assignments, students explore and implement the principles and practices of creativity, innovation, and leadership.

Course Prerequisites

N/A

Special Course Designations

Communication (C) Course

Course Learning Outcomes

- Explain design thinking, its foundations, principles, and main phases
- Define human needs and problems from research
- Identify sources of information outside the classroom that can be helpful for the design problems and technical problems presented in class
- Develop advanced visualization prototyping tools for innovative design products or services.
- Gain skills to advance quality of visual communication and presentation
- Formulate ideas and find solutions through ideation and iteration
- Produce low-fidelity prototypes for design solutions

- Evaluate design outcomes
- Analyze, critique, and present work in a productive and effective manner using appropriate terminology
- Communicate effectively with audience through written and verbal presentations

Textbook and/or Resource Materials

Reading assignments will be given from articles and research papers that are available through the university. There is no required textbook for the course, but the recommended textbook will complement the topics covered in the course.

Course Structure

Students will have both individual and group activities. The individual activities will be in the form of in-class presentations and writing assignments. The group activities will account for developing design solutions for specific social issues.

- **Individual Assignments/Writings: 25%**
- **Individual Presentations: 10%**
- **Quizzes 5%**
- **Team Project: 25%**
- **Individual Project: 25%**
- **Final Paper 10%**
- **Total: 100**

Assignments

Individual Writings

Students will be given written assignments to be submitted on Canvas. Students will also document their findings and research progress using Notion. These assignments will include the items below:

- Analysis of good/bad design examples (400 words)
- Analysis of reading (400 words)
- Research on Prior/existing design works (400 words)
- User study analysis of one user (400 words)
- Reflection on the final project and user study (400 words)

Students will be able to revise their work based on the formative feedback from the instructor.

To receive C credit for this course, you must pass the writing and speaking components.

Individual Presentations

Students will be required to present on an innovative topic of their choosing. The total time for individual students will be a 12 minute presentation during the semester. Students will present these items in their presentation:

- Existing Design analysis and research

- Reading analysis and discussion
- Applications and Results

Students will receive formative feedback on the draft of their presentation before each presentation. There may be other forms of presentations that require teamwork. The team presentations will be graded separately, and they won't be included into the individual grades.

To receive C credit for this course, you must pass the writing and speaking components.

Quizzes

There will be quizzes given via Canvas throughout the semester. These will be on topics covered in class during lectures, readings, and other assigned materials.

Weekly Assignments

There will be multiple weekly assignments that support teams in developing their projects. This will include concept research, ideation, user study, prototyping, and small design exercises to practice specific techniques in design research, user study, ideation, brainstorming, prototyping, implementation, evaluation, and presentation.

Project 1 - Individual Project

Students will individually carry out the first steps in the human centered design process. First the student will pick a TAMU student organization that interests them and perform user research and observations - gathering data and contextual information. Students will then analyze these findings and bring to light - user needs and opportunities. Using this analysis students will share an innovative opportunity statement - that synthesizes their research and analysis. A well formatted presentation and design document will be turned in and presented to the class.

List of Student Organizations: <https://stuactonline.tamu.edu/app/search>

Project 2 - Group Project

Following the individual project, teams will form based on proposals that have similarities in theme. Teams will range from 3-4 individuals. Upon team formation students will work together to pick a particular TAMU organization they want to continue working with. From here students will diverge and brainstorm innovative solutions to the problem statement. The collaborative design process will be the critical point of the activities. Each team project will follow a human centered design process that includes 1) discovering user's needs, 2) analysis, 3) design, 4) prototyping, and 5) implement/deploy.

Teams will be responsible for the timely submission of milestones and final deliverables for each project.

Communication Course Requirements

- **To receive C credit for this course, students must pass the writing and speaking components.**
- Each student must write at least 1250 words related to their research and design projects.
- Each student must present their project and ideas at least for 5 mins.

Grading Policy

To aid in understanding the method by which your project assignments are evaluated, read the following carefully. The actual grading criteria will be related to these categories but will be more specific in nature.

F: 59 and below: *The student work is unresolved; the intentions are unclear and major criteria or goals lack resolution. Presentation is incomplete and/or of poor quality. There is a complete lack of problem solving, creative content and/or technical merit.*

D: 60 -69: *The solution has problems in two or more major areas; skill and design development is marginal or incomplete. The project lacks imagination and/or design potential.*

C: 70 -79: *The student has solved the problem, but the solution lacks depth of understanding; Some program goals are not completely satisfied and the solution contains little promise even though most issues have been addressed.*

B: 80 -89: *The student work shows imagination and potential. Presentation and design content is good; program requirements are fulfilled but in need of more refinement or development. There are no major issues that would require a total redesign of the project.*

A: 90- 100: *The student work has imagination and the solutions to the problems show understanding and thought. The problem solution is highly developed and well presented; the entire assignment shows depth and breadth and is well coordinated. The assignment's potential has been achieved.*

Graded Attendance and Participation

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments. Please refer to [Student Rule 7](#) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Your grade will be based upon attendance and active participation in a wide range of class activities. You are expected to be in class from start to finish every day that we have class, (There are exceptions such as health, please see the University Attendance policy: <https://student-rules.tamu.edu/rule07/>). If you do not come to class (this includes virtual attendance), if you leave class early, or are late to class, you risk losing points on your participation grade. Your attendance is worth 5% of your total grade. It is the student's responsibility to inform the instructor of any intended absences. **A student's final grade will be dropped a letter grade for each unexcused absence after 3 unexcused absences (from 4 unexcused absences).** A student, who attends all classes, is well prepared, contributes to class discussion, and demonstrates their engagement with the material and their peers will earn an A for participation.

Excused absences: A list of excused absences can be found in Student Rule 7.1 (<http://studentrules.tamu.edu/rule07/>). Except for absences due to religious Holidays, the student must notify his or her instructor in writing (acknowledged e-mail message is acceptable) prior to the date of

absence if such notification is feasible. In cases where advance notification is not feasible (e.g., accident, or emergency) the student must provide notification by the end of the second working day after the absence. This notification should include an explanation of why notice could not be sent prior to the class. If the absence is excused, the instructor must either provide the student with an opportunity to make up any quiz, exam or other graded activities or provide a satisfactory alternative to be completed within 30 calendar days from the last day of the absence.

Excused Absences for Religious Holidays: Texas House Bill 256 (effective 9/1/03) states “An institution of higher education shall excuse a student from attending classes or other required activities, including examinations, for the observance of a religious holy day, including travel for that purpose. A student whose absence is excused under this subsection may not be penalized for that absence and shall be allowed to take an examination or complete an assignment from which the student is excused within a reasonable amount of time after the absence.”

Late Work Policy

- **Weekly assignments** will not be accepted late. Failure to show your work in class on the due date will result in a zero. **Other assignments (including projects, documentation, research)** can be submitted late, but with a penalty of 5 points per day, up to 10 days. Later than that will result in a zero.
- Work submitted by a student as makeup work for an excused absence is not considered late work and is exempted from the late work policy. (See [Student Rule 7](#).)
- Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student’s grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.
- Please refer to [Student Rule 7](#) in its entirety for information about makeup work, including definitions, and related documentation and timelines.
- Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor” ([Student Rule 7, Section 7.4.1](#)).
- “The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence” ([Student Rule 7, Section 7.4.2](#)).
- Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24](#).)

Course Schedule

Class attendance is expected. Students should report any known future absences at least one week prior to the absence. Any graded class activities missed (e.g., exams, presentations, homework submissions) during unapproved absences cannot be made up without the instructor’s prior approval and a valid excuse.

Course Topics and Weekly Schedule



Date	Lecture Topics	Lab Topics	Reading	Dues
Week1 Jan. 13 Jan. 15	• Course Introduction • Good/bad design	• Individual Presentations • Design Analysis • Ice Breaker Activity • Introductions	The Design of Everyday Things by Don Norman Chapter 1	
Week2 Jan. 20 Jan. 22	• Overview of design thinking concepts and innovative approaches • Talk about design analysis	• Audience Observation • Project 1 Introduction • Paper Plane Prototyping Activity	The Design of Everyday Things by Don Norman Chapter 6 The Best Paper Airplane - FoldNFly	• Discussion: Analysis of good/bad design
Week3 Jan. 27 Jan. 29	• Introduction of Human centered design • Creativity in Design	• Observation of the target audience and their behaviors • Individual presentations Begin	How Pixar Fosters Collective Creativity Bird by Bird by Anne Lamott	
Week4 Feb. 3 Feb. 5	• Empathy and Research • User Personas	• Making a persona	The Story of Spotify Personas - Spotify Bird by Bird by Anne Lamott	• Discussion: Student Org Observations
Week5 Feb. 10 Feb. 12	• Defining the Question • Making a problem statement	• Making a creative brief	The Goldilocks Approach to Animation Creative Briefs - MOWE Bird by Bird by Anne Lamott	• Quiz
Week6 Feb. 17 Feb. 19	• Project 1 Presentations (Tuesday + Thursday)	• Form Teams • Affinity Diagrams		• Project 1 Due
Week7 Feb. 24 Feb. 26	• Art of Ideating • What is innovation? • Visual Ideation	• Brainstorming and ideation • Sketching and mockups	How Do Designers Overcome Creative Blocks? -	Individual Writing: Analysis of Reading (Don Norman + Anne



			DesignLab	Lamott)
Week8 March 3 March 5	- Peer Evaluation: Getting Feedback on ideas - Cybernetic Systems	Narrowing down ideas	How to ask for design feedback - Webflow Cybernetics in Society and Art	Discussion: Feedback and Cybernetics
SPRING BREAK				
Week9 March 17 March 19	Prototyping	- Paper prototyping - Figma Prototyping - User journey map		
Week10 March 24 March 26	User Testing	Testing Prototypes		- User study plan - Write-up: Prototype feedback
Week11 March 31 April 2	- Innovation in Visualization - Using AI - bring up other areas (e.g., VR)	Worktime	The medium is the message - McLuhan The work of Art in the age of Mechanical Reproduction - Benjamin	Discussion: Work of Art in the age of AI generation
Week12 April 7 April 9 (Michael)	Creating Good Documentation	Worktime		
Week13 April 14 April 16 (Jimin)	Validation/Proof of concept - User Research methods for design/IRB	Worktime		Quiz Make a Trailer
Week14 April 21 April 23	Project Presentations	Project Presentations		Project 2 Due
Week15 April 28	TUESDAY - REDEFINED AS	TUESDAY - REDEFINED AS FRIDAY		Final deliverables due (Make a

	FRIDAY			trailer) • Individual Writing 5: Reflection on the final project and user study
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Optional Course Information Items

Canvas (LMS): Information on Canvas is for the sole use of Texas A&M University students enrolled in the class. This online repository contains materials (images, articles, etc.) copyrighted by others that have been included for the purpose of instruction only, under the "fair use" provision of the U.S. copyright laws. Copying these materials to any other site or use of these materials for any other purpose is prohibited. Please log in at <https://canvas.tamu.edu/>

Technology Support - if you encounter technical difficulties, please contact ITS/Helpdesk for any technical support with your LMS, computer, and software. Contact the Help Desk directly at 979-845-8300. Pipeline technical support e.g. (UV layout is not correct, importing assets in Unreal, etc.) you should contact the instructor and/or the TA by email or on Zoom chat.

Learning Resources –Please seek additional learning resources such as the TAMU University Writing Center or related LinkedIn Learning modules appropriate for the course topics.

- LinkedIn Learning: Sign in by using your TAMU email
https://www.linkedin.com/learning?trk=homepage-learning_nav-header-logo
- Software YouTube Channels: <https://www.youtube.com/user/MayaHowTos>
<https://www.youtube.com/user/UnrealDevelopmentKit>
<https://www.youtube.com/user/Allegorithmic>
- More software tutorials will be uploaded on Class Google drive.

The [Study Hub website](#) also lists many on-campus learning resources to support you in achieving academic excellence.

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

Please refer to [Student Rule 7](#) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to [Student Rule 7](#) in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" ([Student Rule 7, Section 7.4.1](#)).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" ([Student Rule 7, Section 7.4.2](#)).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24](#).)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" ([Section 20.1.2.3, Student Rule 20](#)).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact Disability Resources in the Student Services Building or at (979) 845-1637 or visit disability.tamu.edu. Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see [University Rule 08.01.01.M1](#)):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, you will be able to control how the report is handled, including whether or not to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

Students wishing to discuss concerns in a confidential setting are encouraged to make an appointment with [Counseling and Psychological Services](#) (CAPS).

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University's [Title IX webpage](#).

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors that influence a student's academic success and overall wellbeing. Students are encouraged to engage in proper self-care by utilizing the resources and services available from Counseling & Psychological Services (CAPS). Students who need someone to talk to can call the TAMU Helpline (979-845-2700) from 4:00 p.m. to 8:00 a.m. weekdays and 24 hours on weekends. 24-hour emergency help is also available through the National Suicide Prevention Hotline (800-273-8255) or at suicidepreventionlifeline.org.

College and Department Policies

College and departmental units may establish their own policies and minimum syllabus requirements. As long as these policies and requirements do not contradict the university level requirements, colleges and departments can add them in this section.

Statement of Inclusion

Texas A&M University is committed to enriching the learning and working environment for all visitors, students, faculty, and staff by promoting a culture that embraces inclusion, diversity, equity, and accountability. Diverse perspectives, talents, and identities are vital to accomplishing our mission and living our core values

Care of Facilities

Please respect your facilities (studio space, photo lab, shop, labs, etc.).

The use of spray paint, spray adhesive or other surface-altering materials is not permitted in the Langford Complex, except in designated zones (we do have spray booth facilities located on our complex). Students who violate this rule will be liable for the expenses associated with repairing damaged building finishes and surfaces.

Throughout the semester and at the end of the semester, your area must be clean of all trash. You are responsible for cleaning your studio, desk, and locker at the end of each semester.

No power tools may be used in the design studio. No dust or odor-producing processes may be conducted in the studio. No wet casting processes may be conducted in the studio. The college shop and spray booth facilities must be used for the above-mentioned processes. Professional behavior and conduct is expected of each student.

All studio desks must be covered. In addition, students must have at minimum an 18" x 24" cutting mat at their desk.

Studio Policy (Required of all Studios)

All students, faculty, administration, and staff of the School of Performance, Visualization & Fine Arts at Texas A&M University are dedicated to the principle that the Design Studio is the central component of an effective education. They are equally dedicated to the belief that students and faculty must lead balanced lives and use time wisely, including time outside the design studio, to gain from all aspects of a university education and world experiences. They also believe that design is the integration of many parts, that process is as important as product, and that the act of design and of professional practice is inherently interdisciplinary, requiring active and respectful collaboration with others.

Students and faculty in every design studio will embody the fundamental values of optimism, respect, sharing, engagement, and innovation. Every design studio will therefore encourage the rigorous exploration of ideas, diverse viewpoints, and the integration of all aspects of Visualization (practical, theoretical, scientific, spiritual, and artistic), by providing a safe and supportive environment for thoughtful innovation. Every design studio will increase skills in professional communication, through drawing, modeling, writing, and speaking.